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Diabe-teach: a randomized controlled trial of a gamified approach to enhance medical undergraduates' knowledge and comprehension of diabetes mellitus

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Abstract

The increasing prevalence of diabetes mellitus worldwide necessitates that medical undergraduates acquire a deep understanding of the disease to ensure accurate diagnosis and effective management. Traditional teaching methods, while foundational, often lack the interactive elements that enhance student engagement and knowledge retention. This study aimed to evaluate the effectiveness of a novel educational board game, "Diabe-teach," in enhancing knowledge retention among medical students compared with conventional self-study methods.

A randomized controlled design involving 56 preclinical medical students at Dubai Medical College for Girls was used. The participants were randomly assigned to either the game-based learning group or the control group, with both groups receiving identical instructional materials on diabetes. Knowledge retention was assessed via pre- and post-test scores, whereas student engagement and perceptions were measured via a feedback survey.

The results indicated that while both groups showed significant improvement in post-test scores ($p < .001$), the game-based group demonstrated a notably greater frequency of correct responses and a greater tendency to recognize knowledge gaps, as evidenced by the increased "I do not know" responses compared with incorrect guesses in the control group. Feedback from the game-based learning group highlighted enhanced confidence, improved communication skills, and a preference for the gamified approach over traditional methods. The findings suggest that the "Diabe-teach" board game is an effective tool for teaching complex medical concepts, fostering both knowledge retention and critical self-assessment among students. These results contribute to the growing body of literature on game-based learning, supporting its potential as a valuable adjunct to traditional medical education methods. Further research could explore the application of this approach across other medical disciplines.

Keywords Diabetes mellitus, Educational games, Board games, Active learning, Gamification

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Background and aims

The global surge in diabetes mellitus incidence underscores the critical need for medical undergraduates to acquire a thorough understanding of this disease [8]. Such comprehensive knowledge of diabetes is essential for accurate diagnosis and effective management, as inadequately educated students may struggle to provide optimal care, potentially resulting in complications and adverse health outcomes [16, 34]. Traditional educational methods, including didactic lectures, textbook readings, and clinical case studies, provide foundational knowledge but often lack interactive elements that can enhance engagement and retention [31–34].

Emerging research highlights that integrating interactive tools—both digital and non-digital—into medical curricula can effectively address these limitations, providing students with more engaging and dynamic learning experiences [13, 23]. Among the array of interactive tools, educational board games have demonstrated unique advantages. Board Games not only enhance knowledge retention and critical thinking skills but also foster a collaboration and encourage communication across a wide range of disciplines [3, 17, 19, 24, 30, 34]. Unlike digital simulations or online tools, board games provide a tangible and immersive experience that encourages face-to-face interaction and collaborative learning [1, 3, 34]. This hands-on approach reinforces learning through immediate feedback and iterative play, promoting both strategic thinking and problem-solving in a dynamic and engaging way [31, 33, 34].

While simulations and online tools are valuable for creating scalable and realistic learning environments, they often face challenges such as technological barriers and economic challenges [7, 10, 25]. Board games, on the other hand, are cost effective and excel in their ability to create a social and collaborative atmosphere, where peer interactions drive engagement and deepen understanding [1, 3, 5, 22, 34]. Furthermore, board games can be easily customized to align with specific educational objectives and are practical to implement with minimal technological requirements, making them an accessible tool for educators [5, 22].

Despite their benefits as engaging and interactive learning tools, the widespread adoption of board games is still limited by various practical, pedagogical, and technological challenges. For instance, while board games are engaging for many learners, they may not resonate equally with students who prefer structured or traditional didactic teaching methods [6, 9, 28]. Additionally, designing an effective educational board game to cover complex topics can be time-intensive, and its implementation may require substantial facilitation to ensure alignment with learning outcomes [6, 9, 28]. Logistical challenges, such as resource constraints and competing priorities in packed curricula, can also pose barriers to their widespread adoption [6].

Although previous research highlights the potential of board games in medical education, the majority of studies focus on qualitative outcomes, such as perceived engagement and learner satisfaction, rather than providing robust quantitative evidence of their impact on specific learning outcomes [1, 22]. This study aims to evaluate the impact of a "Diabe-teach" board game-based learning approach on medical undergraduates' knowledge retention concerning diabetes mellitus. By comparing the effectiveness of this method with that of traditional educational approaches, this study seeks to provide quantitative insights into how board games can enrich diabetes education. The evaluation will measure knowledge retention levels post assessment and gauge student engagement with board games versus conventional self-study methods.

The findings contribute to the literature on game-based learning, offering valuable perspectives on its application in teaching complex topics and potentially guiding future educational innovations in other medical disciplines.

Methodology

Ethical approval

This project received approval from the Research and Ethics Committee at Dubai Medical College for Girls (REC No IBR-2022-23-F-17). Before participation, the subjects provided informed consent, and they were explicitly assured of the confidentiality of their information, adhering to ethical considerations throughout the study.

Study design

This study employed a randomized controlled design with consent from preclinical medical undergraduates (Years 1 to 3) from Dubai Medical College for Girls. The preclinical phase was chosen as the stage of the curriculum at which the students were about to take the endocrinology course, ensuring that they had not yet studied the topic in detail. This phase was selected based on its educational significance, as students in preclinical phase are developing foundational knowledge and have not yet been exposed to detailed endocrinology content. This ensures a fair assessment of the impact of the intervention. Additionally, logistical convenience, such as the accessibility of this cohort during the endocrinology course, facilitated their inclusion in the study.

A power analysis was conducted to ensure the study was adequately powered. As a total of 64 students were enrolled in that phase in our institution, the calculation indicated that a minimum of 52 participants (26 per

group) was required to detect a medium to large effect size (d = 0.8) with a significance level of 0.05 and a power of 0.8. With 56 participants (28 in each group), the study met and exceeded the calculated sample size requirements, ensuring sufficient power for statistical analysis.

The participants were randomly assigned to one of two groups: the game-based learning/intervention group or the control group. The participants were randomly assigned to one of two groups: the game-based learning/intervention group or the control group. Randomization was conducted using a computer-generated randomization process via the Randomizer.org (1997–2024). Unique identification numbers were assigned to participants, and the website generated group allocations to ensure an unbiased and systematic distribution.

After undergoing a pretest, all participants were given the same instructional material focusing on the pathophysiology, clinical features, investigation, and management of diabetes. To ensure comparable baseline knowledge across groups, all participants were instructed to study only the material provided before the session. To ensure comparable baseline knowledge across groups, a pretest was conducted for all participants before the intervention. The pretest scores were intended to be compared statistically using an unpaired t-test, with a *p*-value of more than 0.001 considered nonsignificant, to confirm that both groups had similar levels of knowledge.

The game-based learning group engaged in the "Diabe-teach" board game session, whereas the control group participated in a structured self-study hour. The control group was provided with printed handouts and digital materials covering identical content to the board game, focusing on the pathophysiology, clinical features, investigations, and management of diabetes mellitus. The self-study session was organized in a quiet, supervised room to minimize distractions and ensure consistency across participants. Students were instructed to review the materials individually and were not permitted to discuss or collaborate during the session. The process is depicted in Fig. 1.

Inclusion and exclusion criteria

- *Inclusion criteria*: Participants were included if they were in the preclinical phase of the undergraduate medical curriculum, provided written informed consent indicating their willingness to engage in the research, and had not previously studied diabetes in detail.
- *Exclusion criteria*: Individuals were excluded if they dropped out during the study or if they reported having previously studied for diabetes in depth in the consent form. Additionally, students whose pretest scores were identified as outliers (more than 2 standard deviations above the mean) were excluded to control for bias from prior knowledge.

Session timeline

The study followed a structured two-hour session timeline to ensure consistency and reproducibility:

- Pretest (15 min): Participants completed a pretest consisting of 10 MCQs designed to assess baseline knowledge.
- Intervention (75 min): Participants engaged in either the structured self-study session or the game-based learning session (15 min orientation and trial

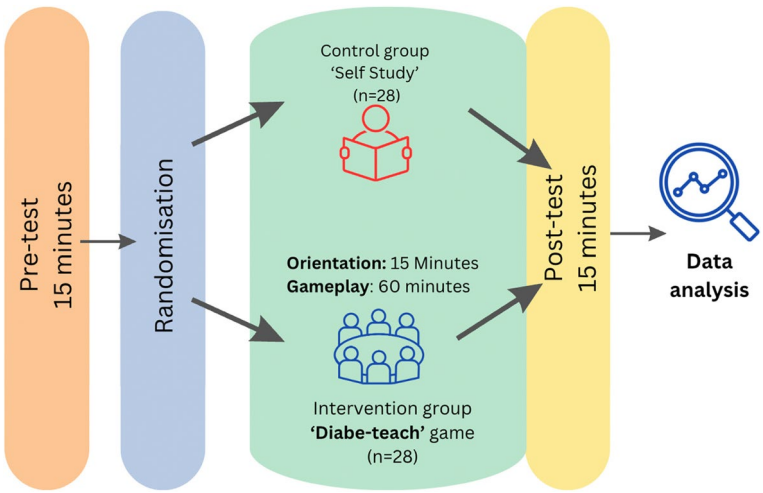


Fig. 1 Steps of the methodology employed for the control and intervention groups (Diabe-teach)

followed by two round of game play each lasting 30 min).

- Post-test (15 min): Immediately following the intervention, participants completed the same 10 MCQs to assess knowledge.
- Feedback Collection (15 min): Participants in the game-based learning group provided feedback via an anonymous online questionnaire designed to capture their perceptions of the gamified approach.

This structured timeline ensured a systematic flow of activities, allowing sufficient time for each phase while maintaining the session's feasibility within two hours.

Game mechanics

The board game, "Diabe-teach," involved 49 cards, each featuring a significant keyword selected by the instructor (Appendix 1). The 49 terms selected for the "Diabe-teach" game ensure coverage of key diabetes concepts while maintaining an optimal level of complexity. The key words were printed on A4-sized paper, and the cards were cut. These cards were shuffled and arranged in a 7×7 grid format on the table before the game began. Each player received paper and a pen to study the cards and construct chains of at least four 'connected' keywords. For a chain to be valid, each keyword had to touch the next keyword either horizontally or diagonally (Fig. 2). The 7×7 grid format allows students to make meaningful connections between four or more terms, enhancing learning and retention without overwhelming them. This balance of educational rigor and playability promotes active engagement and deeper understanding. The participants were required to explain each chain at the end of the round. To ensure that all participants understood the game mechanics, a trial round was conducted before the actual gameplay. During the trial round, students practiced forming chains and explaining their connections, guided by the instructor to address any uncertainties. After the trial round, the cards comprising the game board were reshuffled and rearranged into a new 7×7 grid format to ensure a fair start for the actual game. This ensured that familiarity from the trial round did not provide an advantage during the main gameplay.

Each game session, including the trial round, was timed to ensure consistency. The trial round lasted 15 min, during which participants familiarized themselves with the rules and mechanics. For the actual game, two rounds were played each of approximately 30 min. The cards were shuffled before the each round of play. Gameplay in a round was allocated 30 min, in which during the first 20 min participants analysed the grid, formed chains, and recorded their connections and in last 10 min explained

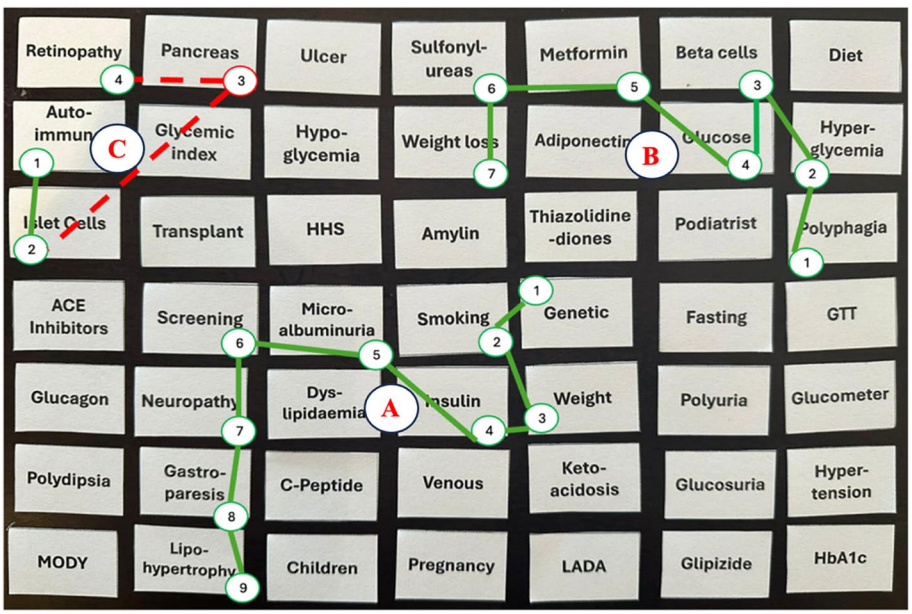


Fig. 2 Examples of a game board and connected chains. The cards were shuffled and arranged in a 7×7 arrangement. Both A and B are valid chains connecting 9 and 7 keywords, whereas C is invalid. *Chain A*: 'Genetics', 'smoking' (and over) 'weight' (people often develop) 'Insulin' (resistance/deficiency that can lead to) 'Microalbuminuria' (which undergo regular) 'screening' (else can develop other complications such as) 'neuropathy', 'gastroparesis', 'lipodistrophy'. *Chain B*: 'Polyphagia' (and) 'hyperglycemia' (are often seen in) 'beta cell' (destruction that regulates) 'glucose' (metabolism). This is managed by drugs such as 'metformin', 'sulfonylurea', and 'weight loss'. *Chain C*: 'Autoimmune' (destruction of) 'islet cells' (in) 'pancreas' (cause diabetes that has complications) and 'retinopathy'. As the keywords 'pancreas' and 'islet cells' do not touch each other, the chain was considered invalid

their chains to their peers and receive feedback from the instructor.

Approval of the connecting chains was subject to consensus among team members, with any ambiguities resolved by the instructor, whose decision was considered final. The winner was determined by the participant with the longest chain, and in cases of ties, the participant with the maximum number of chains was declared the winner.

Data collection

Quantitative data were collected via online pre- and post-assessments consisting of 10 identical multiple-choice questions designed to measure knowledge retention in patients with diabetes mellitus. The multiple-choice questions (MCQs) were categorized at cognitive levels 2 and 3 of Bloom's taxonomy and were carefully reviewed and approved by subject specialists to ensure their relevance and accuracy. The MCQs were reviewed through a structured peer-review process involving three subject specialists with expertise in endocrinology and medical education. Each MCQ was assessed for clarity, clinical relevance, and alignment with the learning objectives of the study. Feedback from the specialists was incorporated to refine question wording, ensure unambiguous distractors, and eliminate potential bias. The revised MCQs underwent a validation study with 10 medical students from a cohort outside the study population to confirm comprehension and alignment with cognitive levels 2 and 3 of Bloom's taxonomy. Each MCQ had 5 options: one correct answer, 3 distractors and one 'I don't know' option to mitigate forced guessing. These responses were recorded separately and excluded from the calculation of test scores, but their frequency was analysed independently to evaluate self-awareness. To assess the effectiveness of the intervention, knowledge retention was measured by comparing pretest and post-test scores within and between groups.

The post-test was administered immediately after the conclusion of the game and self-study sessions to prevent confounding of results due to an extended delay. This timing was chosen to balance immediate recall with short-term retention, providing an objective measure of knowledge gained from the respective interventions.

Feedback collection

Following the post-test, the participants in the game-based learning group completed a feedback survey The questionnaire was designed specifically for this study based on a thorough literature review of gamification in education and was validated by four experts. Later it was piloted with 15 students to ensure clarity and comprehension. No changes were made to the questionnaire following the pilot study, as the pilot participants unanimously confirmed the clarity and relevance of all questions. The survey assessed the perceptions of the students regarding gamification sessions and "Diabe-teach" games through anonymous online surveys. Nine Likert-scale questions were used to measure the perceptions of the students (Appendix 2). A scale of 1–5 was used, with a score of 1 = highly disagree, 2 = disagree, 3 = neutral, 4 = agree, and 5 = highly agree. The questionnaire evaluated various features of the "Diabe-teach" game, such as ease of playing, acceptance, understanding of the content, communication skills, analysis and correlation skills, stress level and comparison with typical small group discussion sessions.

Statistical analysis

For each student in both sessions, the number of correct and incorrect answers from the pretests and post-tests were compared via an unpaired t test ($p < .001$) to determine whether there were significant differences in knowledge retention between the board game-based learning group and the control group. The Statistical Package for the Social Sciences (SPSS) version 26.0 was utilized for the analysis. Before applying the unpaired t-test, the assumptions of normality were assessed using the Kolmogorov–Smirnov test. Only datasets meeting these assumptions were included in the t-test analysis. Additionally, students' feedback on a 5-part Likert scale was analysed categorically for their perceptions of the gamified approach and the percentage of maximum responses for each category was calculated. The internal consistency of all questionnaire items was checked by measuring Cronbach's alpha.

This study aims to contribute to the literature on game-based learning in medical education by providing evidence on the effectiveness of the "Diabe-teach" board game approach. Insights from this study will help determine whether this method enhances knowledge retention, engagement, and satisfaction compared with traditional learning methods in the context of diabetes education.

Results

A total of 56 students participated in the study, with 28 students assigned to each group: the game-based learning group and the control group. The gamification intervention session lasted 1 h and 15 min, including a trial round and two rounds of gameplay. Data collection was done by pretest scores, post-test scores and feedback survey. The normality of the data was assessed via the Kolmogorov–Smirnov test, and the assessment scores are expressed as the means. Comparisons within and between the control

Table 1 Details of the responses of the control and intervention ("Diabe-teach") groups in the pretest and posttest

	Group	Correct answers	Wrong answers	"I don't Know"	Mean pretest	SD
Pretest	Control	25.0%	42.2%	32.8%	25	10
	Intervention	27.9%	48.2%	23.9%	27.9	9.2
Post test	Control	68.9%	25.7%	5.4%	68.9	14.2
	Intervention	81.8%	9.3%	8.9%	81.8	9.8

Table 2 Summary of the statistical analysis of the variables

Variables	Statistical test employed	*p* value	T value	Interpretation
Pretest scores of control and Diabe-teach group	Independent t test	0.27	−1.11	Not significant
Post-test scores of control and Diabe-teach group	Independent t test	< 0.001	−3.93	Significant
Pretest and post-test of Control group	Dependent t test	< 0.001	14.2	Significant
Pretest and post-test of intervention group	Dependent t test	< 0.001	20.8	Significant
Post-test 'I don't know' responses, in of control and interventional group	Independent t test	< 0.001	−7.0	Significant
Post-test Incorrect responses, in of control and interventional group	Independent t test	< 0.001	6.6	Significant

and intervention groups were performed via Student's t test. A p value of < 0.001 was considered significant.

The analysis revealed that the data were normally distributed between the control group (Kolmogorov–Smirnov p value = 0.154) and the intervention group (Kolmogorov–Smirnov p value = 0.210) and both exceeded the threshold for significance ($p > .05$)The mean scores from the pretests and post-tests for the board game-based learning group and the traditional learning group are presented in Table 1. Comparisons within and between two groups were performed via Student's t tests.

Table 2 presents a comprehensive summary of the results from all the statistical analyses. An independent samples t test was conducted to compare the pretest scores between the control group (M = 25.0, SD = 10.0) and the "Diabe-teach" group (M = 27.9, SD = 9.2). The results indicated that there was no statistically significant difference between the two groups, t(df) = −1.11, p = .27, 95% CI [−8.0, 2.2], d = 0.30, suggesting that both groups had comparable baseline knowledge prior to the intervention, ensuring that any differences observed in the post-test can be attributed to the intervention rather than pre-existing disparities. In the post-test, both the "Diabe-teach" game-based learning group (M = 81.8,SD = 9.8) and the self-study group (M = 68.9,SD = 14.2) showed a statistically significant increase in correct responses compared to their pretest scores (t (Randomizer.org. [27]) = 15.21,p < .001 and t (Randomizer.org. [27]) = 15.21,p < .001 respectively). This indicates that knowledge content improved for both traditional and gamified approaches. However, a striking difference was noted in the "I do not know" responses, which were significantly higher ($p < .001$) in the intervention group, whereas choosing wrong answers was significantly more common ($p < .001$) in the self-study group. This finding is particularly important, as it suggests that gamified learning not only enhances students' accuracy but also heightens their awareness of their knowledge gaps. The higher frequency of "I do not know" responses in the intervention group indicates that students were more likely to recognize when they did not fully understand a concept rather than guessing and potentially reinforcing incorrect information. This self-awareness is crucial in medical education, where recognizing the limits of one's knowledge is as important as the knowledge itself.

Figure 3 summarizes the results from the feedback survey assessing students' opinions on the "Diabe-teach" board game. The survey's overall Cronbach's alpha was 0.81, confirming its internal consistency and reliability. Additionally, the Cronbach's alpha for individual items ranged from 0.71 to 0.93, indicating that each item contributed appropriately to the survey's overall reliability. All the students (100%) unanimously agreed that the "Diabe-teach" game significantly boosted their confidence in their ability to discuss complex concepts and enhanced their ability to correlate different pieces of information. When the game-based approach was compared with traditional self-study methods, all the participants (100%) expressed a preference for the "Diabe-teach" game, highlighting that it facilitated a deeper comprehension of the material. None of the participants reported feeling stressed or anxious during the game, and they all recommended using this game in future classes. Nearly 99% of the students found the rules of the game easy to

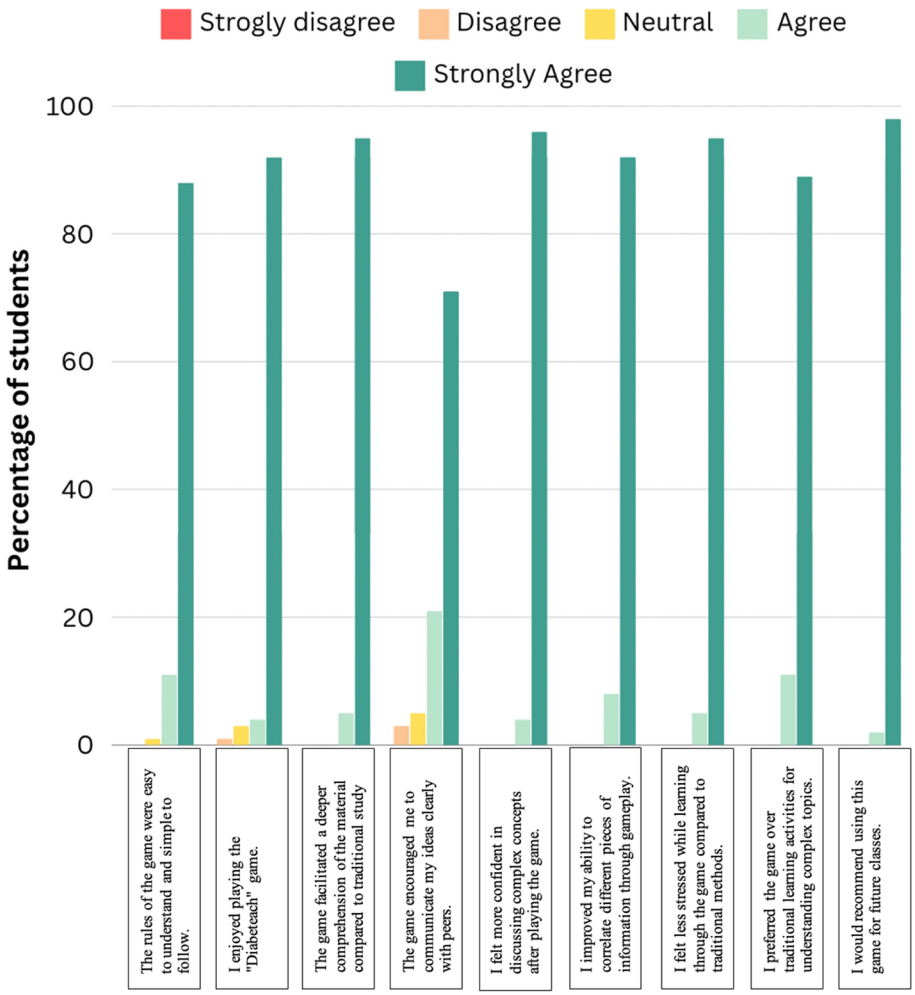


Fig. 3 Results of survey items regarding students' perceptions of the "Diabe-teach" game

understand and simple to follow, whereas over 95% found the gamification experience highly enjoyable. Additionally, almost 92% of the students felt that the game encouraged them to communicate their ideas more clearly with their peers, reinforcing its effectiveness as a tool for collaborative learning.

Discussion

Medical students must develop a robust understanding of diabetes, a complex condition that requires multifaceted management [8]. While traditional educational methods such as didactic lectures and textbook readings provide essential information, they often lack the interactive and engaging elements needed for deeper learning [5, 22]. To meet evolving educational demands, innovative teaching strategies such as gamification have emerged [5, 22]. Gamification has been widely documented for its benefits in enhancing the learning process and improving student engagement [4, 5, 12, 22, 34]. To say that interest in game-based learning has surged recently is an understatement. While digital and video games have dominated much of the discourse, numerous literature reviews [5, 22, 25] have shed light on the significant educational potential of board games. These reviews consistently highlight that board games are not merely recreational activities but also powerful tools that can significantly enhance students' understanding of complex scientific concepts. Unlike digital games, board games offer a tangible, interactive learning experience that can be employed effectively in both formal educational settings and informal learning environments. This study aims to assess the effectiveness of Diabe-teach, a board game specifically designed to foster learning through active, game-based methods, challenging traditional teaching modalities while engaging students in a more dynamic educational process. By evaluating its impact, this study seeks to contribute to the growing body of evidence supporting board games as valuable educational resources.

The comparable pretest scores between the board game-based learning group and the self-study group, with no significant difference ($p = .27$), indicate that both groups started with similar baseline knowledge levels. The significant increase ($p < .001$) in correct responses observed in the post-test for both groups demonstrated that both traditional and gamified approaches were effective in enhancing students' knowledge content. While improvements in players' knowledge through diabetes-related games, as highlighted by Diehl et al. [14], Twist and Ragsdale [34], and Eukel et al. [16], directly support our findings, numerous other studies [5, 25] on board games emphasize the versatility and potential benefits of game-based learning approaches in enhancing knowledge retention and application in medical education. Given that the "Diabe-teach" model is flexible and easy to apply, we anticipate its use across various disciplines.

Another important finding in our study was the striking difference in response patterns between the two study groups, which adds depth to our understanding of these educational strategies (Table 2). In particular, the significantly ($p < .001$) greater frequency of "I do not know" responses in the intervention group contrasts with the greater frequency ($p < .001$). of incorrect answers in the self-study group. This finding suggests that, like previous studies, this gamified learning approach not only improved students' accuracy [16, 34] but also encouraged a more reflective learning process [29]. The willingness of students in the intervention group to admit uncertainty, rather than guessing, may reflect deeper engagement with the material and a more critical assessment of their own knowledge. This aligns with the educational principle that recognizing the limits of one's knowledge is a crucial step in effective learning, particularly in medical education, where the implications of uncertainty can be significant.

These findings have important implications for medical educators. The ability of the "Diabe-teach" game to foster self-awareness in students suggests that gamification could be a valuable tool for promoting metacognitive skills [1, 29], which are essential for lifelong learning and professional development. By encouraging students to identify and address their knowledge gaps, gamified learning can complement traditional methods, potentially leading to more comprehensive and durable learning outcomes. We suggest that future research explore the impact of gamified learning in various contexts and across different medical topics and analyse our findings through longitudinal studies that assess the long-term effects on knowledge retention and clinical application.

The feedback survey results provide compelling evidence for the effectiveness and positive reception of the "Diabe-teach" board game as an educational tool. The strong Cronbach's alpha of 0.81 for the survey, along with high reliability scores for individual items (ranging from 0.71 to 0.93), underscores the robustness of the data collected. The students unanimously reported that the game significantly enhanced their confidence in their ability to discuss complex medical concepts and improved their ability to integrate and correlate information—critical skills in medical education. This unanimous preference for the game over traditional self-study methods suggests that "Diabe-teach" not only made learning more engaging but also deepened students' comprehension of the material. The nontrivial nature of "Diabe-teach" allows for multiple correct answers, reducing anxiety and encouraging active participation [18, 26]. The overwhelmingly positive feedback on the game's ease of use, enjoyability, and ability to foster clear communication among peers highlights its potential as a valuable tool for collaborative learning. This aligns with findings from Curl et al. [11], who reported that their diabetes-related board game improved communication; however, in their study, the focus was on interactions between patients and caregivers, whereas in our case, improvements were noted among students. Furthermore, reviews by Hui and Yunus [20] and Alsawaier [2] underscore the effectiveness of board games in enhancing speaking competence, citing multiple studies that report similar benefits. These parallels reinforce the notion that board games, such as "Diabe teach," can significantly contribute to the development of communication skills in various educational contexts. These findings suggest that "Diabe teach" could be effectively integrated into the medical curriculum to enhance both individual and group learning experiences.

A comparative analysis with similar educational games, such as "Candyland," [34] and diabetes-themed escape rooms [16], highlights the distinct advantages of "Diabe-teach." All the games mentioned above primarily use a trivia format; though, "Candyland" tries to alleviate stress by incorporating a dice-based mechanism [34]. This core reliance on single-answer questions can limit cognitive engagement. In contrast, the open-ended format of "Diabe-teach" emphasizes on constructing meaningful connections between keywords, which fosters critical thinking and a deeper understanding of diabetes. The flexible and nontrivial nature of "Diabe-teach" enhances student engagement by reducing performance pressure [18, 26] and promoting active learning [11, 21]. Moreover, "Diabe-teach" addresses some limitations of traditional board games. For example, while "Candyland" [34] and other trivia-based games may become predictable and repetitive over time, "Diabe-teach" incorporates a dynamic gameboard setup where the arrangement of keywords changes each game due to shuffling. This variability ensures that each session remains engaging and

prevents monotony. Additionally, "Diabe teach" encourages students to analyse and correlate keywords within therapeutic contexts, which simulates real-world clinical reasoning and decision-making processes.

The practical benefits of board games over digital tools are also evident. "Diabe-teach" is cost effective, does not require advanced technological infrastructure, and facilitates face-to-face interaction [15]. The tactile nature of board games enhances spatial reasoning and memory retention [5, 16, 34], aspects that are often less emphasized in digital learning environments. These attributes make board games a versatile and sustainable option for diverse educational settings [5, 16, 34].

Overall, "Diabe-teach" represents a valuable approach in medical education. By incorporating game-based learning, this approach not only improves knowledge retention but also fosters critical thinking, enhanced communication, and collaboration and provides an engaging educational experience. The study highlights the potential of "Diabe-teach" as an effective pedagogical tool, offering a compelling alternative to traditional methods and demonstrating significant advantages over existing educational games such as "Candyland."

Limitations and future directions

While the study of "Diabe-teach" reveals promising results, several limitations warrant attention. The lack of standardized board games in medical education hinders the ability to compare different games and integrate them consistently across institutions. The study's small sample size, demographic homogeneity, and reliance on self-reported feedback limit the generalizability of the findings. Additionally, the short-term assessment of outcomes leaves questions about the game's long-term impact on knowledge retention and clinical application unanswered. Another limitation of this study is the lack of additional analyses to account for potential confounding variables, such as prior academic performance or learning styles. While the randomization process was designed to ensure balanced groups, it is possible that unmeasured factors could have influenced the outcomes. Future studies could include baseline measures of academic performance, such as cumulative GPA, or utilize statistical techniques like regression analysis to control for these variables.

Future research should focus on developing standardized board games that align with medical curricula, allowing for better comparison and integration. Moreover, faculty support and training are essential for addressing concerns about curriculum overload and time constraints when these tools are incorporated. The feasibility of using board games in various educational environments should be explored, including the development of digital versions for interactive learning experiences. Long-term studies are needed to evaluate the impact of board games on examination outcomes, knowledge retention, and overall competence improvement in medical education. While board games such as "Diabe teach" show potential, further research is needed to establish their effectiveness over traditional learning methods.

Conclusion

This study underscores the effectiveness of "Diabe teach," a board game designed to enhance medical students' understanding of diabetes through interactive learning. The significant improvements in knowledge retention and cognitive engagement, as demonstrated by the substantial increase in correct answers and decrease in 'unknown' responses, highlight the potential of the board game as a powerful educational tool. Compared with traditional didactic methods and other educational games such as "Candyland," "Diabe-teach" offers a unique advantage through its non-trivia format, dynamic gameplay, and emphasis on higher-order thinking skills such as the correlation and analysis of data.

To optimize diabetes education at various levels, targeted recommendations are essential. At the individual level, medical students should incorporate gamified tools such as "Diabe-teach" into their self-study routines, as these methods have been proven to enhance knowledge retention and critical thinking skills. Institutions should integrate game-based learning approaches into their curricula, particularly in courses involving complex topics such as diabetes mellitus and provide faculty with training to effectively implement these innovative teaching methods. At the policymaker level, there should be a push to promote the adoption of gamification in medical education standards to improve student engagement and preparedness for clinical practice. Additionally, supporting research on educational tools such as "Diabe-teach" will enable evidence-based curriculum development, ensuring that medical education evolves to meet contemporary needs.

The findings of our study contribute to the growing body of literature on game-based learning and advocate for innovative pedagogical approaches in medical education. Future research should explore the broader applicability of such game-based methods across other medical disciplines and educational contexts, further advancing our understanding of effective teaching strategies.

Looking ahead, the widespread adoption of gamification in medical education has the potential to create a transformative impact by fostering a generation of healthcare professionals who are not only knowledgeable but also skilled in critical thinking, collaboration, and

self-awareness. By integrating these engaging and interactive tools into the curriculum, medical institutions can prepare students to navigate the complexities of modern clinical practice, ultimately improving patient outcomes and shaping the future of global healthcare.

Supplementary Information

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Supplementary Material 1.

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Authors' contributions
MS conceptualized the study design. MS, HI and FHMN analysed the data. MS wrote the first draft of the manuscript. All the authors critically revised the article for important intellectual content and approved the final manuscript.

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Data availability
The datasets used and/or analysed during the current study are available from the corresponding author upon reasonable request.

Declarations

Ethics approval and consent to participate
The Dubai Medical College for Girls Institutional Review Board approved this study (Ref No: IBR2022-23-F-17). Informed consent was obtained from all participants in the study and conformed to the principles outlined in the 1964 Declaration of Helsinki.

Consent for publication
Yes.

Competing interests
The authors declare no competing interests.

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